

AgentPProF: Semantic Profiler for Long Horizon AI Agents

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Abstract

AI agents increasingly orchestrate long-running activities with users, tools, and system resources for days and weeks. To improve quality, safety, and cost efficiency of AI agents, developers need to determine where failures happen, what triggers unsafe effects, and what tasks consume the most budget. They can then optimize those tasks to reduce token and time cost. In systems software, profiling answers similar questions by aggregating resource consumption and attributing it to responsible code paths to identify hotspots. Yet existing agent observability tools focus on per-execution debugging and tracing rather than cross-run, long term profiling, making these questions difficult to answer at scale. Agent observability needs profiling, not only debugging. Profiling agents is challenging: the responsible entities are task intent like *diagnose authentication*, *compare branches* rather than code paths, and lack stable identifiers for aggregation. We propose a *semantic operation stack model* that adapts profiling to agent trajectories. Uniform *operations* represent all activities, and *operation stacks* replace the runtime call stack, enabling hierarchical attribution at different granularities. We observe that an agent’s task occupies a contiguous span and decomposes into subtasks, so we introduce *recursive operation segmentation*, which recursively splits trajectories at task boundaries. AGENTPPROF is a profiler that aggregates agent trajectories into pprof-compatible profiles, enabling flame graph visualization and analysis. On CodeTraceBench, AGENTPPROF reaches 0.764 B³ F1 against human annotations. On three problem-localization benchmarks, the profile raises MAP by up to 56%. AGENTPPROF effectively attributes resources, locates problems, and helps optimize token cost at practical cost. AGENTPPROF is available at <https://github.com/eunomia-bpf/agentsight>.

1 Introduction

AI agents [1, 44, 45] orchestrate multi-step activities that span two layers, high-level intent (prompts, LLM calls, tool invocations) and low-level system effects (process spawns, file and network I/O). As agents evolve from short, scripted workflows into complex systems

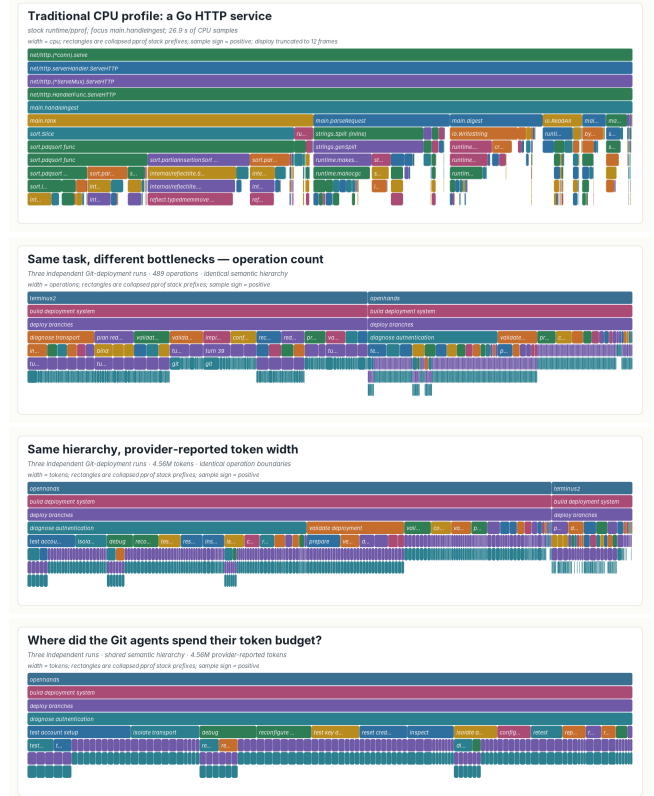


Figure 1: One renderer, four standard pprof profiles. (a) CPU time in a Go HTTP service. (b, c) Three agent executions under one fixed hierarchy: width is operation count (b) or tokens (c). (d) Token profile focused on diagnose authentication.

that run for hours and days, with a large number of interactions

each, production teams accumulate many such trajectories over weeks or months.

To improve agent quality, safety, and cost efficiency, developers need to analyze operational behavior across trajectories and interactions. Which categories of tasks consume the most token budget? Where do failures concentrate across workflows? Which behavioral patterns trigger unsafe system effects? Answering these questions is the first step. Acting on the answers, by optimizing high-cost tasks and fixing failure-prone workflows, is the goal. Answering these questions requires analysis and evaluation across many trajectories [24, 28], a task becoming costly for manual inspection or per-trajectory LLM evaluation [46] at scale. In traditional systems software, profiling answers these questions by aggregating resource consumption and attributing it to responsible entities, distinct from per-execution debugging and tracing.

Yet existing agent tools support debugging and tracing but not profiling. Some [2, 17, 19, 32] organize events into per-execution span trees, effective for diagnosing a single run but unable to aggregate by task intent or workflow phase. Others [9, 16] cluster inputs or extract structured signals, but characterize *input distributions* (“30% of users ask code questions”) rather than *resource attribution* (“review tasks consume 40% of the token budget”), because tags at the request level do not propagate to downstream tool calls and system effects.

Adapting profiling to agent trajectories is challenging. In traditional software, the responsible entities are code paths, and profiling is straightforward because function names are stable and runtime call nesting provides the attribution hierarchy. Agent behavior differs. To answer the questions above, the responsible entities must be task intent and workflow phase, not code paths. Agent trajectories lack the two properties that make traditional profiling possible. Prompts are natural language, so two prompts expressing the same intent share no common string. Agent events have no runtime call-stack hierarchy because prompts, tool calls, and system effects are not nested by execution the way function calls produce stack frames. A sequence of prompts can be a task or multiple tasks, and a task can be part of a bigger task.

Despite these differences, the fundamental profiling methodology transfers to agent trajectories by projecting resources onto responsible entities, assigning stable identifiers, and attributing hierarchically. We propose a *semantic operation stack model* with two components. *Operations* are uniform records with string fields and additive measures that represent all agent activities (prompts, LLM calls, tool invocations, and system effects) without type-specific objects. *Operation stacks* provide hierarchical attribution, replacing the runtime call stack. Choosing different fields changes the attribution granularity: the same operations can be attributed by task category, by execution phase, by session, or by action type, and one fixed hierarchy replays under any additive measure (Figure 1).

We implement this model in AGENTPPROF, an offline profiler that compiles local agent trajectories into pprof-compatible profiles (Figure 2). One algorithm, *recursive operation segmentation*, supplies the missing identifiers. Labeling every prompt individually would be prohibitively expensive, but task structure is recoverable at lower cost: a task occupies a contiguous span of a trajectory and decomposes into contiguous subtasks, so recovering it requires only finding the transitions between them. The algorithm finds where

responsibility changes, names the resulting intervals, and yields short names starting with a verb (like *diagnose authentication*) that fold across sessions. The interface is implementation-neutral.

We evaluate whether AGENTPPROF effectively attributes resources, locates problems, and recovers human-recognizable task structure at practical cost. Across all 405 CodeTraceBench [21] trajectories, recursive segmentation agrees with human stage annotations at 0.764 B³ F1 [4], versus 0.663 for a statistical baseline and 0.541 for raw actions. On three complete localization benchmarks, the profile improves ranking over each benchmark’s own diagnostic, raising MAP by 0.031, 0.107, and 0.117; as a navigation aid, it reduces content to inspect by 12 percentage points at equal ranking quality. A profile-derived repair reduces agent tokens by 19% without degrading task quality, and annotation cost is reduced by 20%.

This paper makes three contributions:

- (1) **Semantic operation stack model.** We identify the missing profiling layer in agent observability and propose a model of uniform operations and query-time operation stacks (Background and Design).
- (2) **System: AGENTPPROF.** A profiler that implements this model, producing pprof-compatible output.
- (3) **Evaluation.** We evaluate profiler output against independent ground-truth annotations that stay hidden from the profiler, on eight public benchmarks and three real trajectory datasets.

2 Background and Motivation

LLMs and AI Agents. A typical agent trajectory interleaves two layers of activity. At the intent layer, the agent receives a user prompt, issues LLM calls, and invokes tools (code execution, web search, file editing). Each tool invocation triggers system effects at the lower layer: process spawns, file reads and writes, and network requests. A single trajectory may contain hundreds of cycles between intent and system effects, and a production team accumulates many trajectories over time.

System Profiling. In traditional software, the profiling pipeline works as follows: a profiler periodically samples execution state, attaches each sample to the current call stack, and merges samples with identical stacks. Flame graphs [14] and pprof [13] call graphs visualize the result, where width or node size represents aggregate cost. Engineers use these profiles to find where CPU or memory budget is spent, then optimize the responsible code paths to reduce cost. These tools support flexible aggregation, but they all assume stable function names and a runtime call stack. Pprof’s *tagroot/tagleaf* options can add label-derived pseudo-frames, but only on top of an existing execution stack that agent trajectories do not have.

A Motivating Case. Three independent agent attempts at one real Git-deployment task all failed to deliver the requested password-authenticated endpoint. The developer’s question is *what went wrong, and is it the same problem across all three runs?* Reading three transcripts totaling 489 operations and 4.6M tokens would take hours. Figure 1 places a conventional CPU profile beside the

semantic profile of those three runs. Both are standard pprof profiles drawn by one renderer and read by identical rules, where width is aggregate cost, a parent contains its children, and identical stacks recorded at different moments fold into one frame. What differs is where the frames come from. A call stack supplies net/http.(*conn).serve at every sample for free, but nothing in an agent trajectory supplies diagnose authentication because the three runs express that intent through different prompts and different shell commands. Once that name is constructed, the profile answers the engineer’s question directly.

The profile answers in one view (Figure 1): all three runs share a diagnose authentication responsibility that consumes 46% of their combined token budget but only 21% of operation count. Expanding the view shows all three executions verifying substitute transport paths without ever establishing the target endpoint. The insight is that the agents kept retrying SSH variants instead of fixing the actual credential issue, and operation count alone would hide this because authentication commands are cheap to issue but expensive to verify. The profile points to where to intervene. Adding early credential validation or limiting retry depth in diagnose authentication could cut token cost for similar tasks.

Challenges for Agent Profiling. Adapting profiling to agent trajectories faces two core challenges. The first is that the responsible entities differ. In traditional software, profiling attributes cost to code paths, but in agent trajectories the relevant entities are task intent and workflow phase. Existing tools [3, 32] do not capture these entities because prompts are natural language: two prompts expressing the same intent may differ, so the profiler cannot aggregate them the way it aggregates identical function names. The second challenge is that agent trajectories have no runtime hierarchy for attribution. In CPU profiling, the call stack determines attribution at every level: malloc’s cost belongs to parse, to process, and to main simultaneously. In AI Agents, sequence of prompts may constitute one task or several, and a task may be part of a larger workflow, but no runtime mechanism marks these boundaries or determines at which granularity to attribute cost.

3 Design

Agent profiling needs the three mechanisms that call stacks provide automatically, but must achieve them without a call stack: **R1** (cross-layer projection): connect system-layer consumption to intent-level categories; **R2** (stable identifiers): derive short, repeatable identifiers from natural-language prompts before folding; **R3** (hierarchical attribution): construct attribution hierarchies from data, not execution nesting.

A view is a triple (φ, σ, w) : a predicate φ selects which operations participate, a stack function $\sigma = [f_1, \dots, f_k]$ maps each operation to its frame sequence, and a weight function w assigns a nonnegative measure. Four built-in views cover common questions: tokens (LLM calls weighted by token count), time (all timed operations weighted by duration), files (file operations weighted by event count), and network (network operations weighted by event count). Operations satisfy R1 by representing intent and system-effect activities in a single record with tag inheritance, so intent tags propagate to the system effects they trigger via AgentSight [47]. *Recursive operation segmentation* satisfies R2 and R3 together by deriving short,

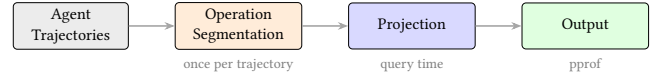


Figure 2: Data-flow pipeline. Local histories and public datasets both produce uniform operations; segmentation runs once, projection and folding at query time.

stable interval names from trajectory content and nesting those intervals into the attribution hierarchy that replaces the runtime call stack. *Operation stacks* then attribute resources at every level of that hierarchy at query time, so the same data supports multiple views without rebuilding the input. The profiling pipeline has four stages: (1) parse raw trajectories into operations, (2) segment trajectories into named nested intervals (or apply mapping rules), (3) project each operation onto a stack frame sequence chosen at query time, and (4) fold operations with identical stacks, summing their weights.

3.1 Semantic Operation Stack Model

This section defines operations and operation stacks, the data structures that satisfy R1 (cross-layer projection) and enable R3 (hierarchical attribution). R2 (stable identifiers) requires deriving identifiers from trajectory content and is addressed in Section 3.2.

Because agent activities span both intent and system-effect layers, a profiler should not distinguish between them in the data representation. AGENTPPROF therefore represents every activity as a single abstraction, the operation, a weighted record with string fields and additive measures. Each operation carries string fields (project, agent, session, prompt_tag, kind, model, path, domain, status, ...) and additive measures (token count, duration, event count). A prompt, an LLM call, a tool invocation, a file read, a GUI click, and a process event are all operations with the same schema, distinguished only by which fields carry values.

An operation stack provides hierarchical attribution for agent trajectories, replacing the runtime call stack. An ordered list of fields $[f_1, f_2, \dots, f_k]$ projects each operation o to a frame sequence $\langle o.f_1, o.f_2, \dots, o.f_k \rangle$, and operations with identical sequences are merged, summing their weights. Unlike a call stack, the fields are chosen at query time, not determined by execution. Changing the fields changes the attribution without changing the data, so the same operations can be attributed by task, by session, or by action type. Sessions and spans are optional fields, so the same data supports both debugging (include session) and aggregate profiling (exclude it).

3.2 Recursive Operation Segmentation

Operation stacks can project any field an operation already carries, but agent trajectories lack the two fields that make traditional profiling work: stable task identifiers and a nesting hierarchy. Labeling every prompt individually would be prohibitively expensive. Recursive operation segmentation derives both fields from trajectory content at lower cost by marking only the transition points where responsibility changes.

The algorithm rests on one structural intuition: an agent’s task occupies a contiguous span of the trajectory, and it decomposes

into smaller contiguous subtasks. We model task structure as a set of nested named intervals, and recovering it only requires finding the transition points between them.

The algorithm reads a trajectory, finds the transition points where the agent’s responsibility changes, and names the intervals on both sides. It then recurses into each interval, cutting again wherever a finer responsibility change remains, and stops when an interval reads as one coherent piece of work. Formally, a trajectory is an ordered step sequence $T = (t_1, \dots, t_n)$, and a segmentation is a set S of named intervals over T that is nested (any two intervals are disjoint or one contains the other), covers every step, and contains the session-wide interval as its root. The algorithm computes S by one recursive rule: `SEGMENT( $I$ )` selects zero or more transition points inside interval I , which partition I into consecutive child intervals. Each child receives a short name starting with a verb and is segmented recursively. Selecting no transition terminates the branch. A step’s operation path is the name chain of the intervals containing it, from the session root to its innermost interval. Equal paths fold across sessions. Concretely, in one Git-deployment execution the algorithm cuts the session into `build deployment system` and, inside it, cuts again where the agent stops writing configuration and starts debugging access, yielding `deploy branches` and `diagnose authentication`. A step inside the latter carries the path `build deployment system > diagnose authentication`, and because the same names reappear in the other two executions, their costs fold together in Figure 1. Any implementation able to find transitions and name intervals (an agent, a language model, or a statistical rule) can serve as the segmenter. In evaluation, we use Codex [31] with GPT-5.6-sol-high, with one fixed instruction per trajectory:

Input: one trajectory (each step’s prompt, command, and output summary, with no labels or scores).
Action: read it, emit a mark wherever the responsibility changes, and use AGENTPPROF to check and update the segmentation, and re-read and update the marks iteratively until you think the segmentation is complete.
Output: sparse path marks, e.g.,
 step 1: [deploy git server]
 step 4: [deploy git server, diagnose SSH access]
 step 15: [deploy git server, verify web endpoint]
 —one line only where the path changes; names start with a verb, one to three words.

The sparse marks still produce a complete segmentation because steps between two marks inherit the preceding path. Segmentation is iterative, so the agent can revise marks incrementally without re-annotating the entire trajectory. The model supports both offline evaluation and online annotation during execution.

4 Implementation

This section describes how AGENTPPROF realizes the design from Section 3. AGENTPPROF is implemented as a Rust CLI (~9.8K LOC, Figure 2). It reads Codex and Claude Code local JSONL history files and AgentSight [47] recordings for system effects. These files are written incrementally during agent execution, so AGENTPPROF can build profiles while the agent is still running. The pipeline has two stages: (1) preprocessing extracts a readable trajectory summary,

and (2) the segmentation agent reads this summary and writes interval marks to an annotation file, which it can revise until satisfied. The CLI validates that marks are nested and cover every step, then emits a standard pprof protobuf profile that `go tool pprof` reads directly. Projection is deterministic: each operation contributes its semantic path and LLM/tool evidence, with session identities kept as pprof labels rather than visible frames, so equal semantic prefixes fold across sessions. Switching the additive measure replays the same annotations and changes only stack widths.

Fields already recorded literally (e.g., skill invocations) become stack levels without annotation cost. Multi-agent orchestration works automatically because subagent operations inherit the outer agent’s semantic path. A non-LLM statistical segmenter is also available, scoring transitions by NPMI [6] and calibrating an unsupervised cutoff with k -means [27].

5 Evaluation

We evaluate whether AGENTPPROF effectively attributes resources, locates problems, and recovers human-recognizable task structure at practical cost. The evaluation uses three data classes. *Real inputs:* 41 long-horizon coding-agent sessions (three of which repeat one Git-deployment task and form the RQ1 case), 440 mixed web-agent trajectories [24], and one developer’s complete local session history from the authors’ workstation (1,394 sessions), whose 42 long-horizon development sessions are the annotated portion. *Annotated trajectories:* 405 CodeTraceBench trajectories with 20,866 operations and 2,948 human-annotated stages [21], 287 OSWorld-Human sessions [42], 1,012 AgentBoard goals [25], and 2,737 published action labels [7]. *Problem-localization benchmarks:* the complete AgentProcessBench, HINTBench, and TraceElephant workloads, with independently annotated faulty operations, over 27,346 operations [8, 11, 40]. All annotations, outcomes, and scores stay hidden from every method until its output is produced. Benchmark trajectories average 8–52 operations (short to medium). The 42-session portion from the research of this paper supplies long-horizon coverage, whose longest sessions span tens of hours, and the RQ4 union covers 27,765 operations.

5.1 RQ1: Does Semantic Profiling Improve Resource Attribution?

Setup. If semantic profiling improves resource attribution, then the same fixed hierarchy should (1) reunite cross-run responsibility that alternative organizations scatter, and (2) reveal materially different bottlenecks when the additive measure changes. We test both on 41 real long-horizon agent trajectories (3,146 user turns, 5,750 operations), including three independent executions of one Git-deployment task whose 735 steps receive 96 recursive annotations reaching semantic depth five.

The motivating case revisited. The Git-deployment case from Section 2 demonstrates that semantic organization reunites cross-run responsibility that alternative organizations scatter. Here we add a third measure (elapsed time) to show measure-sensitivity. The same hierarchy replays under all three measures with exact conservation, and the `diagnose authentication` subtree’s share rises from 21% (count) through 37% (time) to 46% (tokens). Changing only the measure moves the attributed share by more than a factor

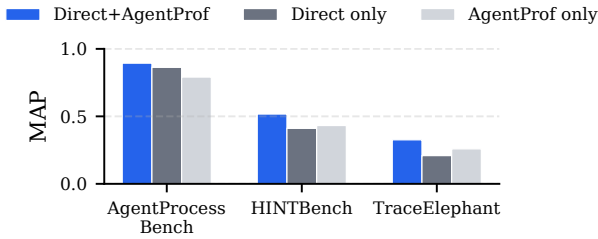


Figure 4: MAP over the complete RQ2 datasets (614, 400, and 220 queries included in MAP). Higher is better.

revealed. We report per-query AP averaged over 614, 400, and 220 queries included in MAP. The 522 trajectories without an annotated faulty operation are consumed for dataset coverage but excluded from MAP. The *direct diagnostic* is the per-operation output of each benchmark’s own judge, including LLM-as-judge methods; it reads the full trace and reference answer before naming the responsible agent and decisive step. *Direct-only* ranks by that diagnostic alone; *Direct+AGENTPPROF* breaks ties by group score (vote mean for AgentProcessBench; 95% Wilson bound otherwise), with HINTBench field order fixed on 80 validation trajectories; *AGENTPPROF-only* uses only the profiler without the benchmark diagnostic (Figure 4).

Results. Direct+AGENTPPROF beats Direct-only by 0.031, 0.107, and 0.117 MAP on the three workloads (all statistically significant; Figure 4). Grouping with retained evidence drives the localization gain. Semantic naming enables cross-run attribution (RQ1) and concentrates the reader’s attention, reducing evidence opened from 65% to 53% (measured below).

Profile-guided reading on TraceElephant. On 220 queries, a Grok 4.5 [43] agent-as-judge ranks operations by likely fault. Full-trace reading reaches MAP .502 at 12.6K tokens/query. A profile-guided reader selects at most five groups and reaches .455 while opening 53% of the source, versus .465 and 65% with raw-action names. A per-query full read is also bounded by the model context window. **Takeaway.** Failure structure surfaced by the profile matches independent human labels across 435 trajectories. Profiles add localization signal on top of existing judges, and semantic names concentrate the agent’s attention, reducing source characters opened from 65% to 53%.

Use case 4: guiding a real repair. The preceding experiments establish correspondence and inspection value; this case tests whether a profile finding can guide an effective repair. A standard AGENTPPROF profile over eight ToolSandbox scenarios isolates a recurring call-ID syntax failure (5/21 tool operations). A profile-only analyst—receiving only the pprof output, not raw traces—identifies the compatibility-layer boundary as the repair target. The derived one-line fix removes all syntax failures and, on 23 held-out confirmation scenarios (69 BEFORE/repair pairs), reduces agent-model tokens by 19.0% while passing a fixed -0.05 quality threshold on official similarity.

Method	B ³ P	B ³ R	B ³ F1	Bound. F1
Codex segmentation	0.793	0.736	0.764	0.480
Statistical recurrence	0.782	0.575	0.663	0.266
Causal Qwen2.5-3B task stack	0.557	0.792	0.650	0.257
Raw-action grouping	0.891	0.388	0.541	—
Native source tree	0.975	0.249	0.397	0.259
Source-native step	0.983	0.221	0.361	0.246

Table 1: Agreement with human stages on all 405 Code-TraceBench trajectories.

5.3 RQ3: How Accurate Are the Identifiers?

Setup. To evaluate whether recursive segmentation recovers task structure that humans would recognize, we compare segmentation output against 2,948 human-annotated stages over all 405 Code-TraceBench trajectories. F1 [4] asks whether operations that belong together end up grouped together; exact adjacent-boundary F1 [36] asks whether cuts land exactly where humans cut. Baselines receive identical inputs (Table 1). Each method output is evaluated at the level it predicts. Literal names use accuracy and macro-F1 [20], permutation-invariant partitions by V-measure [35] or ordinary B³, and adjacent interval boundaries by exact precision, recall, and F1. For legacy field-valued datasets a conversion step maps each predicted group into the same interval-annotation format before AGENTPPROF folds the original additive weights. Codex [31] receives each trajectory (prompts, commands, and output summaries, with no labels or scores visible) and emits sparse interval marks over 17,148 steps without access to stage annotations (loaded after all 405 outputs were fixed): 4,496 marks at semantic depth one (3), two (2,873), three (1,588), or four (32), with no prompt requesting a specific depth. Name normalization then maps the 3,895 free-form interval names to 783 stable short names (e.g., diagnose authentication) with zero adjacent display-path collisions, rejecting unresolved ones. Identical names across sessions merge so that identical paths fold in the final profile.

Results. Codex segmentation reaches 0.764 B³ F1 and 0.480 boundary F1, beating the strongest automatic baseline (multi-resolution recurrence) by +0.101 and +0.214. A stateful per-turn Qwen2.5-3B baseline reaches 0.650 B³ F1, showing smaller models can also segment meaningfully. The marks conserve exactly 20,866 operations and 494,862,929 tokens. Predicted groups remain largely pure subsets of gold stages (B³ precision 0.793), and the residual error is over-segmentation rather than missed transitions: when segmentation disagrees with human stages, it subdivides work rather than merging unrelated responsibilities, preserving per-group coherence. The interface generalizes across method families: on OSWorld-Human [42], a supervised Naive Bayes predictor reaches 0.816 B³ F1, and a no-LLM recurrence segmenter reaches 0.786. The framework supports different segmentation scales: on OSWorld-Human, the unchanged Codex instruction uses coarser task-level boundaries (0.448 B³ F1), while a supervised adapter matched to its finer action-group granularity reaches 0.816. For closed-label literal tags, a Qwen3.6-27B [33] classifier labels 1,012 AgentBoard goals [25] at

0.695 macro-F1, and 2,737 action labels from AutoCodeRover, OpenHands, and RepairAgent trajectories [7, 37] at 0.498 macro-F1. *Takeaway.* Recursive segmentation recovers human-recognizable task structure: 0.764 B³ F1 against human stages, beating the strongest automatic baseline by +0.101. When segmentation disagrees with humans, it predominantly over-segments (subdivides work) rather than merging unrelated responsibilities, preserving per-group coherence. Non-LLM methods implementing the same interface remain viable when no model is available.

5.4 RQ4: What Is the Profiling Cost?

Setup. To evaluate whether profiling cost is practical, we measure segmentation time and tokens and profile replay speed. Cost separates into a one-time automatic segmentation pass per corpus and deterministic construction/replay afterwards.

Results. Segmenting all 405 CodeTraceBench trajectories with Codex (GPT-5.6-sol-high) completes in 37 minutes with up to four workers, averaging 29,754 input and 573 output tokens per trajectory. With marks fixed, construction scales linearly: the 27,765-operation union completes in 1.16 s at 465 MiB peak RSS, adding only 5.25 MiB (1.14%) over raw-action grouping. Deterministic construction of the full 440 trajectories takes 0.26 s for operations and 0.25 s for tokens, so construction cost is dominated by the segmentation step while construction stays sub-second.

Reducing annotation cost. Automatic annotation dominates profile construction. Showing only a compact skeleton plus selected full results (SELECTIVE) instead of every turn’s complete content (FULL) reduces provider tokens by 20.4% on 32 held-out CodeTraceBench task clusters while retaining 100% operation coverage and meeting a fixed -0.03 quality threshold on B³ and boundary F1.

Takeaway. Segmentation is a one-time pass (37 min for 405 trajectories); every later view, measure switch, and inspection costs about one second. Selective evidence reduces annotation tokens by 20.4% with complete coverage and maintained structural quality.

6 Related Work

Classic profilers fold runtime call stacks over stable code identity, and Pivot Tracing groups causally related measurements [13, 14, 26]. Agent observability platforms such as LangSmith, Langfuse, and Phoenix [2, 17, 19] record spans with metadata for per-execution debugging. Analytics layers such as Datadog Patterns and LangSmith Insights [10, 18] roll up cross-trace hierarchies or input distributions, but none constructs recursive semantic responsibility with conserved additive measures in a standard profile. Cross-run analyses (TraceProbe, Graphectory, Act-onomy, CHIEF, Hodoscope, TraceGraph) [12, 22, 30, 38, 41, 48] and per-run diagnosis and localization [5, 8, 11, 15, 21, 23, 28, 29, 39, 40] answer what an agent did and which step went wrong. AGENTPPROF instead asks how much of an additive resource a task or workflow phase is responsible for. It recovers variable-depth responsibility from source content (where Act-onomy fixes three levels and TraceProbe one), conserves measures exactly across count, token, and time widths, keeps LLM/tool calls as leaf nodes, and emits standard pprof. No compared system provides all four, and AGENTPPROF stays complementary to per-run diagnosis, as RQ2 measures directly.

7 Conclusion

Agent observability needs profiling, not only debugging. AGENTPPROF shows that the semantic operation stack model, driven by recursive operation segmentation, brings hierarchical attribution to agent trajectories. Against independent annotations, segmentation recovers human stage structure at 0.764 B³ F1, profiles add localization signal on three complete public benchmarks, and one hierarchy replays across additive measures with exact conservation. A repair guided by a single profile reduces agent-side model tokens by 19.0% without material official-quality loss; selective annotation reduces provider tokens by 20.4% with complete coverage and maintained structural quality. Profilers and debuggers are complementary. The profile answers cross-run questions and guides inspection to the content worth opening, reducing opened content by 12 percentage points while reaching similar ranking. Multi-project evaluation and tracing-ecosystem integration are the most impactful next steps.

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